

Laboratory Report 3: Membrane Transport

Due: Wednesday, December 9 at 11:59 PM

Value: 100 points

Format: Individual scientific report

Purpose

Estimate the isotonic sucrose concentration for potato tissue and explain mass change using water potential and membrane selectivity.

Your task

- Calculate percent mass change for every replicate using $(\text{final} - \text{initial}) / \text{initial} \times 100$.
- Report mean and range for each sucrose concentration and plot mean percent change against concentration.
- Estimate the isotonic concentration by identifying or interpolating the x-intercept; state your method.
- Explain the direction of water movement without saying that sucrose 'pulls' water as a force.
- Evaluate tissue variation, blotting, solution preparation, and balance precision; propose one improved control.

Provided evidence or data

Sucrose	Rep 1 %	Rep 2 %	Rep 3 %
0.0 M	+15.2	+13.8	+14.6
0.2 M	+8.4	+7.9	+8.8
0.4 M	+1.5	+0.8	+1.1
0.6 M	-6.7	-7.3	-6.1
0.8 M	-13.4	-12.8	-14.1

Submit

- 1,200-1,500 word scientific report PDF.
- Appendix with replicate calculations and x-intercept estimate.

Grading criteria

Criterion	Pts	Full-credit evidence
Question and methods	15	Frames the investigation and describes the relevant procedure accurately.
Results	25	Calculations, figures, captions, and statistical summaries are correct and transparent.
Interpretation	30	Explains results using an appropriate cellular mechanism and addresses uncertainty.
Evaluation	20	Identifies limitations, controls, and a justified follow-up experiment.
Scientific form	10	Clear structure, accurate units, citations, and reproducible data presentation.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.